

NOVEMBER 2021 (1)

Question	Answer	Marks
3(a)(i)	one mark for sample, one mark for reason max four Normal Sample any positive value with three decimal places e.g. 5.682 Reason to test that normal data is accepted and processed correctly Erroneous Sample any value that would be rejected e.g. 5.6 or -1.345 or seven Reason to test that erroneous data is rejected	4
3(a)(ii)	Reason to test that 0.000 / -0.001 / highest possible non-positive is rejected and 0.001 / 0.000 / lowest positive number is accepted Sample 1 0.000 Sample 2 0.001	3
3(b)	One mark To check that values are entered as intended // to prevent incorrect values that meet the validation criteria being accepted Two marks Asking the user to enter the value twice and comparing the values // double entry (1) only accepting a value if both entries are identical (1) or Displaying the value as it is entered (1) so the user can put right errors have been made as the value was entered (1)	3

Question	Answer	Marks																																																																	
4	One mark for each correct column <table><tr><th>Counter</th><th>Distinction</th><th>Mark</th><th>Award</th><th>OUTPUT</th></tr><tr><td>0</td><td>0</td><td></td><td></td><td></td></tr><tr><td>1</td><td>1</td><td>88</td><td></td><td></td></tr><tr><td>2</td><td></td><td>74</td><td></td><td></td></tr><tr><td>3</td><td></td><td>60</td><td></td><td></td></tr><tr><td>4</td><td>2</td><td>90</td><td></td><td></td></tr><tr><td>5</td><td>3</td><td>84</td><td></td><td></td></tr><tr><td>6</td><td>4</td><td>87</td><td></td><td></td></tr><tr><td>7</td><td>5</td><td>95</td><td></td><td></td></tr><tr><td>8</td><td></td><td>72</td><td></td><td></td></tr><tr><td>9</td><td>6</td><td>84</td><td></td><td></td></tr><tr><td>10</td><td></td><td>66</td><td></td><td></td></tr><tr><td></td><td></td><td>-1</td><td>0.6</td><td>Highly Commended</td></tr></table>	Counter	Distinction	Mark	Award	OUTPUT	0	0				1	1	88			2		74			3		60			4	2	90			5	3	84			6	4	87			7	5	95			8		72			9	6	84			10		66					-1	0.6	Highly Commended	5
Counter	Distinction	Mark	Award	OUTPUT																																																															
0	0																																																																		
1	1	88																																																																	
2		74																																																																	
3		60																																																																	
4	2	90																																																																	
5	3	84																																																																	
6	4	87																																																																	
7	5	95																																																																	
8		72																																																																	
9	6	84																																																																	
10		66																																																																	
		-1	0.6	Highly Commended																																																															

NOVEMBER 2021 (2)

Question	Answer	Marks																				
	Section B																					
2	One mark for two correct rows Two marks for three correct rows Three marks for four correct rows. <table><tr><th>Statement</th><th>Validation (✓)</th><th>Verification (✓)</th><th>Neither (✓)</th></tr><tr><td>a check where data is re-entered to make sure no errors have been introduced during data entry</td><td></td><td>✓</td><td></td></tr><tr><td>an automatic check to make sure the data entered has the correct number of characters</td><td>✓</td><td></td><td></td></tr><tr><td>a check to make sure the data entered is sensible</td><td>✓</td><td></td><td></td></tr><tr><td>a check to make sure the data entered is correct</td><td></td><td></td><td>✓</td></tr></table>	Statement	Validation (✓)	Verification (✓)	Neither (✓)	a check where data is re-entered to make sure no errors have been introduced during data entry		✓		an automatic check to make sure the data entered has the correct number of characters	✓			a check to make sure the data entered is sensible	✓			a check to make sure the data entered is correct			✓	3
Statement	Validation (✓)	Verification (✓)	Neither (✓)																			
a check where data is re-entered to make sure no errors have been introduced during data entry		✓																				
an automatic check to make sure the data entered has the correct number of characters	✓																					
a check to make sure the data entered is sensible	✓																					
a check to make sure the data entered is correct			✓																			

Question	Answer	Marks
3	One mark per bullet point Normal test data • Test data e.g. 50 (allow any number between 1 and 100 inclusive) • Reason Data that is within range and should be accepted Extreme test data • Test data 100 / 1 • Reason Data at the maximum / minimum end of the range and should be accepted Erroneous test data • Test data e.g. 300 (allow anything that isn't between 1 and 100 inclusive, including other data types) • Reason Data outside the range that should be rejected	6

Question	Answer	Marks																																																							
5	One mark for each correct column <table><tr><th>List</th><th>Value</th><th>List1</th><th>List2</th><th>OUTPUT</th></tr><tr><td></td><td></td><td>0</td><td>0</td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>77</td><td></td><td>77</td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>16</td><td></td><td>93</td><td></td></tr><tr><td>1</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>35</td><td>35</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>-7</td><td></td><td>86</td><td></td></tr><tr><td>5</td><td></td><td></td><td></td><td></td></tr></table>	List	Value	List1	List2	OUTPUT			0	0		2						77		77		2						16		93		1						35	35			2						-7		86		5					5
List	Value	List1	List2	OUTPUT																																																					
		0	0																																																						
2																																																									
	77		77																																																						
2																																																									
	16		93																																																						
1																																																									
	35	35																																																							
2																																																									
	-7		86																																																						
5																																																									

Question	Answer					Marks
5	List	Value	List1	List2	OUTPUT	
		18			Input Error	
	1					
		11	46			
	1					
		12	58			
	2					
		20		106		
	-1				List 1 = 58	
					List 2 = 106	
					List 2 is greatest	

NOVEMBER 2021 (3)

Question	Answer	Marks
3(a)(i)	one mark for sample, one mark for reason max four Normal Sample Reason any password with at least 8 characters and one special character e.g. Password! to test that normal data is accepted and processed correctly Erroneous Sample Reason any value that would be rejected e.g. secret to test that erroneous data is rejected	4
3(a)(ii)	Reason to test that Secret? which has 7 characters is rejected and Secret?? which has 8 characters is accepted Boundary Sample 1 – Secret? Boundary Sample 2 – Secret??	3

Question	Answer	Marks
3(b)	Any two methods, one mark for method, one mark for description max four E.g. Asking the user to enter the password twice and comparing the values (1) only accepting the data if both entries are identical (1) Displaying the password as it is entered (1) so the user can put right errors have been made as the password was typed (1)	4

FOUR MARKS

Question	Answer	Marks																																																																	
4	One mark for each correct column <table><tr><th>Counter</th><th>Pass</th><th>Mark</th><th>Help</th><th>OUTPUT</th></tr><tr><td>0</td><td>0</td><td></td><td></td><td></td></tr><tr><td>1</td><td>1</td><td>88</td><td></td><td></td></tr><tr><td>2</td><td></td><td>24</td><td></td><td></td></tr><tr><td>3</td><td>2</td><td>60</td><td></td><td></td></tr><tr><td>4</td><td></td><td>30</td><td></td><td></td></tr><tr><td>5</td><td>3</td><td>44</td><td></td><td></td></tr><tr><td>6</td><td></td><td>17</td><td></td><td></td></tr><tr><td>7</td><td></td><td>25</td><td></td><td></td></tr><tr><td>8</td><td></td><td>22</td><td></td><td></td></tr><tr><td>9</td><td>4</td><td>54</td><td></td><td></td></tr><tr><td>10</td><td></td><td>6</td><td></td><td></td></tr><tr><td></td><td></td><td>999</td><td>0.4</td><td>Extra Help</td></tr></table>	Counter	Pass	Mark	Help	OUTPUT	0	0				1	1	88			2		24			3	2	60			4		30			5	3	44			6		17			7		25			8		22			9	4	54			10		6					999	0.4	Extra Help	5
Counter	Pass	Mark	Help	OUTPUT																																																															
0	0																																																																		
1	1	88																																																																	
2		24																																																																	
3	2	60																																																																	
4		30																																																																	
5	3	44																																																																	
6		17																																																																	
7		25																																																																	
8		22																																																																	
9	4	54																																																																	
10		6																																																																	
		999	0.4	Extra Help																																																															

Question	Answer	Marks
3	One mark for each correct <u>single</u> line from the validation check, max four Validation check length check check digit range check type check Description checks that the data input is between two values checks that the data input is an integer checks that the data input has three digits checks that the data has been input checks that the data input has the correct digits	4

Question	Answer	Marks																																																				
4(a)	One mark each for columns Number and OUTPUT Two marks for column C first four values (1) last three values (1) Two marks for column D first six values (1) last four values (1) <table><tr><th>Number</th><th>C</th><th>D</th><th>OUTPUT</th></tr><tr><td>7</td><td>0</td><td>3</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td></td><td>1</td><td></td></tr><tr><td>6</td><td>0</td><td>3</td><td></td></tr><tr><td></td><td>1</td><td>2</td><td></td></tr><tr><td></td><td>2</td><td>1</td><td>2</td></tr><tr><td>5</td><td>0</td><td>2</td><td></td></tr><tr><td></td><td></td><td>1</td><td></td></tr><tr><td>4</td><td>0</td><td>2</td><td></td></tr><tr><td></td><td>1</td><td>1</td><td>1</td></tr><tr><td>-1</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table>	Number	C	D	OUTPUT	7	0	3				2				1		6	0	3			1	2			2	1	2	5	0	2				1		4	0	2			1	1	1	-1								6
Number	C	D	OUTPUT																																																			
7	0	3																																																				
		2																																																				
		1																																																				
6	0	3																																																				
	1	2																																																				
	2	1	2																																																			
5	0	2																																																				
		1																																																				
4	0	2																																																				
	1	1	1																																																			
-1																																																						
4(b)	One mark per mark point, max two - to count the factors / the numbers that go into (other than 1 or itself) of a number - to output the number of factors	2																																																				

Question	Answer	Marks
4(c)(i)	One mark per mark point, max two - the value of <u>D</u> becomes zero - division by zero error - endless loop	2
4(c)(ii)	One mark per mark point, max two - after the decision box to test if the number is -1 - insert another decision box to test if the number is less than 4 / less than or equal to 3 - return to INPUT Number if true	2

Question	Answer	Marks																																									
Section B																																											
2	Four marks for five correct rows Three marks for four correct rows Two marks for three correct rows One marks for two correct rows <table><tr><th rowspan="2">Description</th><th colspan="5">Data type</th></tr><tr><th>Boolean</th><th>Char</th><th>Integer</th><th>Real</th><th>String</th></tr><tr><td>a single character from the keyboard</td><td></td><td>✓</td><td></td><td></td><td></td></tr><tr><td>multiple characters from the keyboard</td><td></td><td></td><td></td><td></td><td>✓</td></tr><tr><td>only one of two possible values</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>only whole numbers</td><td></td><td></td><td>✓</td><td></td><td></td></tr><tr><td>any number</td><td></td><td></td><td></td><td>✓</td><td></td></tr></table>	Description	Data type					Boolean	Char	Integer	Real	String	a single character from the keyboard		✓				multiple characters from the keyboard					✓	only one of two possible values	✓					only whole numbers			✓			any number				✓		4
Description	Data type																																										
	Boolean	Char	Integer	Real	String																																						
a single character from the keyboard		✓																																									
multiple characters from the keyboard					✓																																						
only one of two possible values	✓																																										
only whole numbers			✓																																								
any number				✓																																							

Question	Answer	Marks
3	One mark per mark point, max four - Normal test data computerscience@cambridge.org.uk - Reason this is a valid email address (containing the @ symbol) and should be accepted - Erroneous test data computerscienceisgreat - Reason this is just a string, and should be rejected (as an email address needs a single '@')	4

Question	Answer	Marks
4(a)	One mark per mark point, max six MP1 input box MP2 correct check of Score MP3 assign Score to Pass correctly MP4 assign Score to Fail correctly MP5 increment both array counters MP6 correct check of number of scores <pre> graph TD START([START]) --> PC0[PassCount ← 0] PC0 --> FC0[FailCount ← 0] FC0 --> C0[Count ← 0] C0 --> INPUT[/INPUT
Score/] INPUT --> IS50{IS Score ≥ 50?} IS50 -- Yes --> P1[Pass[PassCount] ← Score] P1 --> PC1[PassCount ← PassCount + 1] IS50 -- No --> F1[Fail[FailCount] ← Score] F1 --> FC1[FailCount ← FailCount + 1] PC1 --> C1[Count ← Count + 1] FC1 --> C1 C1 --> IS60{IS Count = 60?} IS60 -- No --> INPUT IS60 -- Yes --> END([END]) </pre>	6

Question	Answer	Marks																																																																														
5(a)	One mark per mark point, max five MP1 correct Counter and Limit columns MP2 correct Value column MP3 correct First column MP4 correct Last column MP5 correct OUTPUT <table><tr><th>Counter</th><th>Value</th><th>First</th><th>Last</th><th>Limit</th><th>OUTPUT</th></tr><tr><td></td><td></td><td>0</td><td>0</td><td>8</td><td></td></tr><tr><td>1</td><td>66</td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>606</td><td>6</td><td>6</td><td></td><td>606</td></tr><tr><td>3</td><td>6226</td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>8448</td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>642</td><td>6</td><td>2</td><td></td><td></td></tr><tr><td>6</td><td>747</td><td>7</td><td>7</td><td></td><td>747</td></tr><tr><td>7</td><td>77</td><td></td><td></td><td></td><td></td></tr><tr><td>8</td><td>121</td><td>1</td><td>1</td><td></td><td>121</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Counter	Value	First	Last	Limit	OUTPUT			0	0	8		1	66					2	606	6	6		606	3	6226					4	8448					5	642	6	2			6	747	7	7		747	7	77					8	121	1	1		121																			5
Counter	Value	First	Last	Limit	OUTPUT																																																																											
		0	0	8																																																																												
1	66																																																																															
2	606	6	6		606																																																																											
3	6226																																																																															
4	8448																																																																															
5	642	6	2																																																																													
6	747	7	7		747																																																																											
7	77																																																																															
8	121	1	1		121																																																																											
Question	Answer	Marks																																																																														
5(b)	One mark per mark point, max two • checks for / outputs 3-digit numbers • ... where the first and last digit are the same	2																																																																														

JUNE 2022 (2)

Question	Answer	Marks																																																																																											
4(a)	One mark for each correct column	7																																																																																											
	<table><tr><th>Counter</th><th>Hot</th><th>Cold</th><th>Serve</th><th>Temp</th><th>Error</th><th>OUTPUT</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td>1</td><td>75</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td>2</td><td>78</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td>3</td><td>84</td><td></td><td></td></tr><tr><td>4</td><td>1</td><td></td><td></td><td>87</td><td></td><td>Too Hot</td></tr><tr><td>5</td><td>2</td><td></td><td></td><td>91</td><td></td><td>Too Hot</td></tr><tr><td>6</td><td></td><td></td><td>4</td><td>80</td><td></td><td></td></tr><tr><td>7</td><td></td><td></td><td>5</td><td>75</td><td></td><td></td></tr><tr><td>8</td><td></td><td></td><td>6</td><td>70</td><td></td><td></td></tr><tr><td>9</td><td></td><td></td><td>7</td><td>65</td><td></td><td></td></tr><tr><td>10</td><td></td><td>1</td><td></td><td>62</td><td></td><td>Too Cold</td></tr><tr><td></td><td></td><td></td><td></td><td>-1</td><td>30</td><td>30</td></tr></table>		Counter	Hot	Cold	Serve	Temp	Error	OUTPUT	0	0	0	0				1			1	75			2			2	78			3			3	84			4	1			87		Too Hot	5	2			91		Too Hot	6			4	80			7			5	75			8			6	70			9			7	65			10		1		62		Too Cold					-1	30	30
	Counter		Hot	Cold	Serve	Temp	Error	OUTPUT																																																																																					
	0		0	0	0																																																																																								
	1				1	75																																																																																							
	2				2	78																																																																																							
	3				3	84																																																																																							
	4		1			87		Too Hot																																																																																					
	5		2			91		Too Hot																																																																																					
	6				4	80																																																																																							
	7				5	75																																																																																							
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	9				7	65																																																																																							
	10			1		62		Too Cold																																																																																					
				-1	30	30																																																																																							
Question	Answer	Marks																																																																																											
4(b)	include a message to explain the value output / e. g. "The percentage of meals not served" // outputting Hot, Cold and Serve	1																																																																																											
4(c)	updating the Serve variable // <code>Serve ← Serve + 1</code>	1																																																																																											

Question	Answer	Marks																													
Section B																															
2	One mark per row, max four <table><tr><th rowspan="2">Description</th><th colspan="4">Types of test data</th></tr><tr><th>Boundary</th><th>Erroneous / Abnormal</th><th>Extreme</th><th>Normal</th></tr><tr><td>test data that is always on the limit of acceptability</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>test data that is either on the limit of acceptability or test data that is just outside the limit of acceptability</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>test data that will always be rejected</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>test data that is within the limits of acceptability</td><td></td><td></td><td>✓</td><td>✓</td></tr></table>	Description	Types of test data				Boundary	Erroneous / Abnormal	Extreme	Normal	test data that is always on the limit of acceptability			✓		test data that is either on the limit of acceptability or test data that is just outside the limit of acceptability	✓				test data that will always be rejected		✓			test data that is within the limits of acceptability			✓	✓	4
Description	Types of test data																														
	Boundary	Erroneous / Abnormal	Extreme	Normal																											
test data that is always on the limit of acceptability			✓																												
test data that is either on the limit of acceptability or test data that is just outside the limit of acceptability	✓																														
test data that will always be rejected		✓																													
test data that is within the limits of acceptability			✓	✓																											

Question	Answer	Marks
3	One mark per mark point, max four - variables are used to represent values that can change during the execution of a program // variables can be used to store the results of calculations / counting / totalling // can store values entered by the user - variable example – any data that is input into a program such as a date - constants represent values that must stay the same throughout the execution of a program - constant example – any value that does not change, such as Pi in mathematical formulae	4

Question	Answer	Marks																																																																																																																																																																											
4(a)	One mark per mark point, max seven MP1 correct In column MP2 correct Logic column MP3 correct Test column MP4 correct Number column MP5 correct Store[Count] column MP6 correct Count and Limit columns MP7 correct Out and OUTPUT columns <table><thead><tr><th>In</th><th>Logic</th><th>Test</th><th>Number</th><th>Store [Count]</th><th>Count</th><th>Limit</th><th>Out</th><th>OUTPUT</th></tr></thead><tbody><tr><td></td><td></td><td></td><td></td><td></td><td>0</td><td>5</td><td></td><td></td></tr><tr><td>1</td><td>TRUE</td><td>2</td><td>9</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>FALSE</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>TRUE</td><td>2</td><td>5</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>3</td><td></td><td>5</td><td>1</td><td></td><td></td><td></td></tr><tr><td>3</td><td>TRUE</td><td>2</td><td>8</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>FALSE</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>TRUE</td><td>2</td><td>10</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>FALSE</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>TRUE</td><td>2</td><td>7</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>3</td><td></td><td>7</td><td>2</td><td></td><td>0</td><td>5</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>7</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></tbody></table>	In	Logic	Test	Number	Store [Count]	Count	Limit	Out	OUTPUT						0	5			1	TRUE	2	9								3								FALSE								2	TRUE	2	5								3		5	1				3	TRUE	2	8							FALSE								4	TRUE	2	10							FALSE								5	TRUE	2	7								3		7	2		0	5								1	7																																														7
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Question	Answer	Marks
4(b)	One mark per mark point, max two - to find / output prime numbers ... store prime numbers in an array	2
4(c)	One mark per mark point, max three MP1 insert a WHILE loop ... // pre-condition loop MP2 ... after Input Number MP3 ... with a condition to enter the loop <code>Number < 3</code> MP4 an error message included within the loop to ask for a re-entry of Number MP5 ...with another input prompt for Number MP6 ENDWHILE closes the loop and the program carries on from REPEAT in the original algorithm OR One mark per mark point, max three MP1 insert a REPEAT loop ... // post-condition loop MP2 ... before Input Number MP3 a conditional statement should be placed after Input Number MP4 ...to check if <code>Number < 3</code> MP5 if the number entered is <code><3</code>, an error message included within the loop to ask for a re-entry of Number MP6 UNTIL <code>Number >= 3</code> closes the loop and the program carries on from REPEAT in the original algorithm	3

Question	Answer	Marks
5	One mark per mark point, max six MP1 input box MP2 correct totalling using Total MP3 correct counting using Counter MP4 correct conditional statement for Counter MP5 correct calculation of Average MP6 correct outputs of Total and Average <pre> graph TD START([START]) --> InitTotal[Total ← 0] InitTotal --> InitCounter[Counter ← 0] InitCounter --> Input[/INPUT Number/] Input --> AddTotal[Total ← Total + Number] AddTotal --> IncCounter[Counter ← Counter + 1] IncCounter --> Decision{IS Counter = 100?} Decision -- No --> Input Decision -- Yes --> CalcAvg[Average ← Total / Counter] CalcAvg --> OutputTotal[/OUTPUT Total/] OutputTotal --> OutputAvg[/OUTPUT Average/] OutputAvg --> END([END]) </pre>	6

Question	Answer	Marks										
2(a)	One mark for each single correct line from a description to statement <table><thead><tr><th>Description</th><th>Pseudocode statement</th></tr></thead><tbody><tr><td>a statement to count</td><td>FOR Count ← 1 TO 10</td></tr><tr><td>a statement to total</td><td>Value ← Value + NewValue</td></tr><tr><td>a statement to start a pre-condition loop</td><td>WHILE Value > 10 DO</td></tr><tr><td>a statement to start a post-condition loop</td><td>REPEAT</td></tr></tbody></table>	Description	Pseudocode statement	a statement to count	FOR Count ← 1 TO 10	a statement to total	Value ← Value + NewValue	a statement to start a pre-condition loop	WHILE Value > 10 DO	a statement to start a post-condition loop	REPEAT	4
Description	Pseudocode statement											
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a statement to start a pre-condition loop	WHILE Value > 10 DO											
a statement to start a post-condition loop	REPEAT											

Question	Answer	Marks
2(b)	One mark for each point: • Initialisation of total to zero before loop • appropriate loop controls • totalling statement inside the loop, must use array Number[] with an index • calculation of average outside loop • output of average outside loop Example: <pre> Total ← 0 FOR Count ← 1 TO 50 Total ← Total + Number[Count] NEXT Count Average ← Total / 50 OUTPUT "The average is ", Average </pre>	5

Question	Answer	Marks
3	One mark for each point max two. • check that the program works as expected • check for logic/runtime errors • check that the program rejects any invalid data that is input • check that the program only accepts reasonable data One mark for example Normal // erroneous // abnormal // extreme // boundary	3

Question	Answer	Marks
5	One mark for a suitable hierarchical structure One mark for suitable names for the sub systems for user input and display options One mark for sub systems for user inputs, (choice of display,) food order and payment One mark for sub systems for display output types, pictures and list For example: <pre> graph TD A[Food ordering system] --> B[User Input] A --> C[Display options] B --> D[Display choice] B --> E[Food order] B --> F[Payment] C --> G[Pictures] C --> H[List] </pre>	4

Question	Answer	Marks
6(a)	One mark for each error identified and correction - Line 05 OUTPUT UsefulEnergyOut should be INPUT UsefulEnergyOut - Line 06 IF TotalEnergyIn <> -1 AND UsefulEnergy <> -1 should be: IF TotalEnergyIn <> -1 AND UsefulEnergyOut <> -1 - Line 11 UNTIL TotalEnergyIn <> -1 OR UsefulEnergyOut <> -1 should be: UNTIL TotalEnergyIn = -1 OR UsefulEnergyOut = -1	3
6(b)	One mark for checking for >= 92 One mark for outputting "A-rated" only if the condition is met For example <pre>IF Efficiency >= 92 THEN OUTPUT "A-rated" ENDIF</pre>	2

Question	Answer	Marks																																													
8(a)	<table><tr><th>NumberSales</th><th>Total</th><th>SaleValue</th><th>Average</th><th>OUTPUT</th></tr><tr><td>0</td><td>0</td><td></td><td></td><td></td></tr><tr><td>1</td><td>5.50</td><td>5.50</td><td></td><td></td></tr><tr><td>2</td><td>8.90</td><td>3.40</td><td></td><td></td></tr><tr><td>3</td><td>15.15</td><td>6.25</td><td></td><td></td></tr><tr><td>4</td><td>19.00</td><td>3.85</td><td></td><td></td></tr><tr><td>5</td><td>8.00</td><td>-11.00</td><td></td><td></td></tr><tr><td></td><td></td><td>0</td><td>1.6</td><td>Average sale value 1.6</td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table> One mark for each column NumberSales, Total, SaleValue One mark for columns Average and OUTPUT	NumberSales	Total	SaleValue	Average	OUTPUT	0	0				1	5.50	5.50			2	8.90	3.40			3	15.15	6.25			4	19.00	3.85			5	8.00	-11.00					0	1.6	Average sale value 1.6						4
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8(b)	Error – one mark for identification of error for example including negative numbers / not differentiation between negative and positive values Correction One mark for correct placement and one mark for appropriate action For example – after the input box insert a decision box to reject negative numbers	3																																													

Question	Answer	Marks
1(a)	One mark for each correct line. Program development life cycle description develop an algorithm to solve the problem using structure diagrams, flowcharts or pseudocode detect and fix the errors in the program identify the problem and its requirements write and implement the instructions to solve the problem Program development life cycle stage analysis coding design evaluation testing	4
1(b)	One mark for naming or describing each component part, max three For example: inputs // what is put into the system processes // actions taken to achieve a result outputs // what is taken out of the system storage // what needs to be kept for future use	3

Question	Answer	Marks
3(a)	One mark per mark point, max two - Validation is an automated check carried out by a computer ... to make sure the data entered is sensible/acceptable/reasonable	2
3(b)	One mark for each appropriate test data, max three One mark for each correct accompanying reason, max three For example: Normal – 75 Reason – the data lies within the required range and should be accepted Abnormal – Sixty Reason – this is the wrong data type and should be rejected Extreme – 200 Reason – the highest value in the required range that should be accepted	6

Question	Answer	Marks
5(a)	One mark per mark point, max four - Line 04 / IF Number < 0 should be IF Number > 0 - Line 10 / Exit ← 1 // Line 01/ Exit ← 1 and Line 02 / WHILE Exit <> 0 should be Exit ← 0 // should be Exit ← 0 and WHILE Exit = 0 - Line 13 / ENDIF should be ENDWHILE - Line 14 / OUTPUT "The total value of your numbers is ", Number should be OUTPUT "The total value of your numbers is ", Total Correct algorithm: <pre> 01 Exit ← 1 02 WHILE Exit <> 0 DO 03 INPUT Number 04 IF Number > 0 05 THEN 06 Total ← Total + Number 07 ELSE 08 IF Number = 0 09 THEN 10 Exit ← 0 11 ENDIF 12 ENDIF 13 ENDWHILE 14 OUTPUT "The total value of your numbers is ", Total </pre>	4

Question	Answer	Marks
5(b)	One mark per mark point, max four - Initialise a new (counting) variable ... Count ← 0 // to count the acceptable numbers - Insert a counting statement between lines 05 and 07 ... Count ← Count + 1 - Add a new output after the loop/after line 13 / at the end (of the program) ... OUTPUT Count	4

Question	Answer	Marks																																																												
7(a)	One mark per correct column, max four <table><tr><th>Pointer</th><th>Letter</th><th>Choice</th><th>OUTPUT</th></tr><tr><td>1</td><td>F</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td></tr><tr><td>5</td><td></td><td></td><td></td></tr><tr><td>6</td><td></td><td></td><td>Letter F is represented by Foxtrot</td></tr><tr><td></td><td></td><td></td><td>Another Letter? (Y or N)</td></tr><tr><td></td><td></td><td>Y</td><td></td></tr><tr><td>1</td><td>D</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td>Letter D is represented by Delta</td></tr><tr><td></td><td></td><td></td><td>Another Letter? (Y or N)</td></tr><tr><td></td><td></td><td>N</td><td></td></tr></table>	Pointer	Letter	Choice	OUTPUT	1	F			2				3				4				5				6			Letter F is represented by Foxtrot				Another Letter? (Y or N)			Y		1	D			2				3				4			Letter D is represented by Delta				Another Letter? (Y or N)			N		4
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7(b)	(Linear) search	1																																																												

JUNE 2023 (2)

Question	Answer	Marks
1	A	1

Question	Answer	Marks
3	One mark for each correct answer • structure diagram / chart • flowchart • pseudocode	3

Question	Answer	Marks
4(a)	One mark for each point (max three). • range check with acceptable values is (greater than) zero and less than 1000 • presence check to ensure the program will not continue until a value has been entered • type/character check to ensure that a number is entered • length check to ensure there are no more than 3 digits entered	3
4(b)(i)	To verify the data / for verification / as a verification check // to make sure that no changes are made to the data on entry	1
4(b)(ii)	One mark for each point (max three). • use of iteration • use of two inputs • to check that the two inputs are the same / different • use of the given variable Measurement For example <pre> REPEAT OUTPUT "Please enter measurement " INPUT Measurement OUTPUT "Please re-enter measurement " INPUT MeasurementCheck UNTIL Measurement = MeasurementCheck </pre>	3

Question	Answer	Marks
7(a)	• 07 • 04/12 or 16/18 • 02/20	3
7(b)	One mark for each error identified and correction • Line 07 <code>Total ← Total + Number * Counter</code> should be <code>Total ← Total + Number[Counter] * Counter</code> • Line 08 <code>IF Number[Counter] = 0</code> should be <code>IF Number[Counter] = -1 // should be IF Number[Counter] < 0</code> • Line 16 <code>FOR Counter ← 0 TO 5</code> should be <code>FOR Counter ← 1 TO 5</code>	3

Question	Answer	Marks
7(c)	One mark for place in algorithm (max one) • around lines 05 and 06 • line 07 • (immediately) after the input of the number Three marks pseudocode One mark for each point (max three) • Use of REPEAT ... UNTIL // any working loop structure • check for >0 // >=0 • check for <10 // >9 • check for whole number • check for -1 • check for length of digit <> 1 Example <pre> REPEAT OUTPUT "Enter a digit " INPUT Number[Counter] UNTIL Number[Counter] = Round(Number[Counter],0) AND ((Number[Counter] = -1) OR (Number[Counter] > 0 AND Number[Counter] < 10)) </pre>	4

Question	Answer	Marks																																																																																																								
9(a)	One mark for each column F, C and T Two marks for columns X[1] to X[5] all entries correct or One mark for columns X[1] to X[5] with one error <table><tr><th>F</th><th>C</th><th>X[1]</th><th>X[2]</th><th>X[3]</th><th>X[4]</th><th>X[5]</th><th>T</th></tr><tr><td></td><td></td><td>10</td><td>1</td><td>5</td><td>7</td><td>11</td><td></td></tr><tr><td>0</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td>10</td></tr><tr><td>1</td><td>2</td><td>1</td><td>10</td><td></td><td></td><td></td><td>10</td></tr><tr><td>1</td><td>3</td><td></td><td>5</td><td>10</td><td></td><td></td><td>10</td></tr><tr><td>1</td><td>4</td><td></td><td></td><td>7</td><td>10</td><td></td><td></td></tr><tr><td></td><td>5</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>4</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>5</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	F	C	X[1]	X[2]	X[3]	X[4]	X[5]	T			10	1	5	7	11		0	1						10	1	2	1	10				10	1	3		5	10			10	1	4			7	10				5							0	1								2								3								4								5															5
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9(b)	One mark for each point • (bubble) sort data in array • in ascending order	2																																																																																																								

JUNE 2023 (3)

Question	Answer	Marks
4(a)	One mark per mark point, max two - To ensure that data has been accurately copied // to ensure that changes have not been made to the values originally intended when data is copied ... from one source to another	2

Question	Answer	Marks
4(b)	One mark for each appropriate verification check, max two One mark for each correct accompanying use, max two For example: Verification check 1 – Visual check Use – the user looks through the data that has been entered and confirms that no changes have been made. Verification check 2 – Double data entry Use – data is entered twice, the two entries are compared and if they do not match, a re-entry is requested.	4

Question	Answer	Marks
5(a)	One mark for each correct line. Description to check that the data entered is an integer to check that some data has been entered to check that the data entered has an appropriate number of characters to check that an identification number contains no errors Check check digit format check length check presence check type check	4

Question	Answer	Marks
5(b)	One mark per mark point, max three - appropriate REPEAT / WHILE loop begin and end - input of Length - appropriate input prompt / error message - correct loop exit/entry condition / selection Example answers: WHILE Loop <pre> OUTPUT "Enter a number between 15 and 35 inclusive" INPUT Length WHILE Length <15 OR Length > 35 (DO) OUTPUT "Your number must be between 15 and 35 inclusive" INPUT Length ENDWHILE </pre> REPEAT Loop <pre> REPEAT OUTPUT "Enter a number between 15 and 35 inclusive" INPUT Length UNTIL Length >= 15 AND LENGTH <= 35 </pre>	3

Question	Answer	Marks
6(a)	One mark per mark point, max four - Line 01 / Counter $\leftarrow$ 100 should be Counter $\leftarrow$ 0 - Line 03 / While Counter > 100 DO should be While Counter < 100 DO - Line 07 / Total $\leftarrow$ Total + Counter should be Total $\leftarrow$ Total + Number - Line 09 / ENDCASE should be ENDIF Correct algorithm <pre> 01 Counter $\leftarrow$ 0 02 Total $\leftarrow$ 0 03 WHILE Counter < 100 DO 04 INPUT Number 05 IF Number > 0 06 THEN 07 Total $\leftarrow$ Total + Number 08 Counter $\leftarrow$ Counter + 1 09 ENDIF 10 ENDWHILE 11 OUTPUT "The total value of your numbers is ", Total 12 OUTPUT "The average value of your numbers is ", Total / 100 </pre>	4

Question	Answer	Marks
6(b)	One mark per mark point, max five MP1 replace line 03 MP2 with FOR MP3 ... with limits 0 to 99 / 1 to 100 MP4 replace line 05 to check if Number is not positive MP5 ... (if Number is not positive) insert a validation and re-input routine between lines 06 and 07 ... MP6 ... that will repeat until a positive value is entered MP7 remove the counter update / line 08 MP8 replace line 10 / ENDWHILE with NEXT	5

Question	Answer	Marks																																																																													
7(a)	One mark per mark point, max six • correct <code>Total</code> column • correct <code>Value</code> column • correct <code>Five1</code> column • correct <code>Five2</code> column • correct <code>Ten1</code> and <code>Ten2</code> columns • correct <code>OUTPUT</code> column <table><tr><th>Total</th><th>Value</th><th>Five1</th><th>Five2</th><th>Ten1</th><th>Ten2</th><th>OUTPUT</th></tr><tr><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>5</td><td>1</td><td>1</td><td>0</td><td>0.5</td><td>Rejected</td></tr><tr><td></td><td>50</td><td>10</td><td>10</td><td>5</td><td>5</td><td></td></tr><tr><td>50</td><td>52</td><td>10</td><td>10.4</td><td></td><td></td><td>Rejected</td></tr><tr><td></td><td>555</td><td>111</td><td>111</td><td>55</td><td>55.5</td><td>Rejected</td></tr><tr><td></td><td>57</td><td>11</td><td>11.4</td><td></td><td></td><td>Rejected</td></tr><tr><td></td><td>500</td><td>100</td><td>100</td><td>50</td><td>50</td><td></td></tr><tr><td>550</td><td>−1</td><td></td><td></td><td></td><td></td><td>550</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Total	Value	Five1	Five2	Ten1	Ten2	OUTPUT	0								5	1	1	0	0.5	Rejected		50	10	10	5	5		50	52	10	10.4			Rejected		555	111	111	55	55.5	Rejected		57	11	11.4			Rejected		500	100	100	50	50		550	−1					550															6
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7(b)	One mark per mark point, max two • to find if an input is divisible by (both 5 and) 10 • ... add them together and output the total	2																																																																													

NOVEMBER 2023

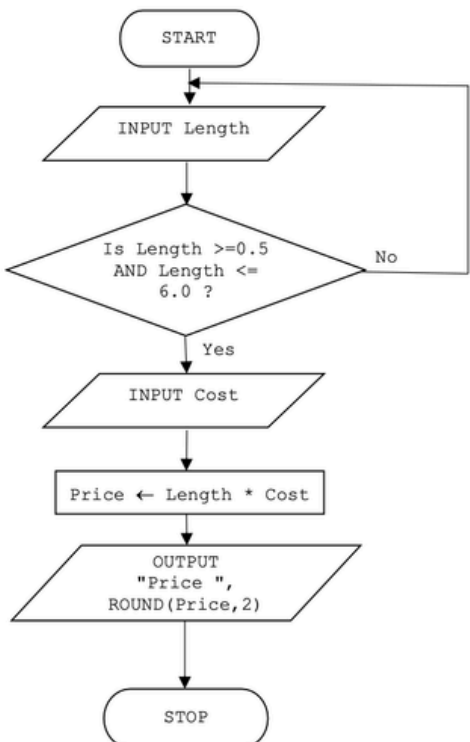
Question	Answer	Marks
4	One mark for each correct word • array • constant • variable	3

Question	Answer	Marks
5(a)	One mark for each point (max two) • simplifying the problem • removing unnecessary details from the problem // selecting elements required • filtering out irrelevant characteristics from those elements	2
5(b)	One mark for each point (max three) • inputs • processes • outputs • storage	3
5(c)	One mark for stage, one mark for matching description (max two) • design (1) details of solution set out (1) • coding (1) program is developed (1) • testing (1) program is tested for errors (1)	2

Question	Answer	Marks
6(a)	Displaying/sort 10 names in alphabetical order 9	1

Question	Answer	Marks
6(b)	One mark for each point (max four) • Initialisation • inputting 10 names • storing the names in an array • sorting the names in alphabetical order using a bubble sort • displaying the 10 names • iteration	4
6(c)	One mark for a meaningful identifier for the array A Names // ArrayNames Two marks for 3 meaningful identifiers for variables One marks for 1 or 2 meaningful identifiers for variables T Temp C Counter L Length	3
6(d)	One mark for each point (max two) • use of comments • use of procedures/functions • use of white space	2

Question	Answer	Marks
8(a)	Range check	1

Question	Answer	Marks
8(b)	 <pre> graph TD START([START]) --> INPUT_Length[/INPUT Length/] INPUT_Length --> DECISION{Is Length >=0.5 AND Length <= 6.0 ?} DECISION -- No --> INPUT_Length DECISION -- Yes --> INPUT_Cost[/INPUT Cost/] INPUT_Cost --> PROCESS[Price ← Length * Cost] PROCESS --> OUTPUT[/OUTPUT "Price ", ROUND(Price,2)/] OUTPUT --> STOP([STOP]) </pre> One mark for each of the following points • correct use of flowchart symbols • working flow lines and complete • both inputs correct • working range check • working calculation • correct output rounded to two decimal places	6
8(c)	One mark for set of test data, one mark for purpose (max four) Example: 1 and 1 (1) normal data to ensure the algorithm accepts this test data (1) -1 and 1 (1) abnormal data for length to ensure that it is rejected (1)	4

Question	Answer	Marks				
8(d)	One mark for two correct headings Two marks for three correct headings Three marks for all headings correct and no other headings unless used in 8(b) <table><tr><td>Length</td><td>Cost</td><td>Price</td><td>OUTPUT</td></tr></table>	Length	Cost	Price	OUTPUT	3
Length	Cost	Price	OUTPUT			
8(e)	One mark for each point (max two) • validate Cost ... • ... with a range/presence check • add another validation check for Length	2				

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Question	Answer	Marks
1	A	1

Question	Answer	Marks
2(a)	Format check	1
2(b)	One mark for each appropriate test data, max two One mark for each correct accompanying reason, max two For example: Normal – 30/12/1960 ... Reason – ... (the date is written in the correct format and) should be accepted. Abnormal – 30/Dec/1960 ... Reason – ... (the month is not written in the correct format and) should be rejected.	4
2(c)	One mark per mark point, max two MP1 check that there are 10 characters in total MP2 if the date is too long/short it will be rejected	2

Question	Answer	Marks												
3(a)	One mark for each correct line. <table><thead><tr><th>Pseudocode statement</th><th>Pseudocode use</th></tr></thead><tbody><tr><td>CALL Colour (NewColour)</td><td>counting</td></tr><tr><td>Value ← (A1 + A2 + A3) / 3</td><td>finding an average</td></tr><tr><td>Loop1 ← Loop1 + 1</td><td>totalling</td></tr><tr><td>IF Count > 7 THEN X1 ← 0</td><td>using a conditional statement</td></tr><tr><td></td><td>using a procedure</td></tr></tbody></table>	Pseudocode statement	Pseudocode use	CALL Colour (NewColour)	counting	Value ← (A1 + A2 + A3) / 3	finding an average	Loop1 ← Loop1 + 1	totalling	IF Count > 7 THEN X1 ← 0	using a conditional statement		using a procedure	4
Pseudocode statement	Pseudocode use													
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Loop1 ← Loop1 + 1	totalling													
IF Count > 7 THEN X1 ← 0	using a conditional statement													
	using a procedure													

Question	Answer	Marks
3(b)	One mark per mark point, max four MP1 initialise a variable to store the lowest number set to a high value (at least 100) / first value in the array MP2 loop structure to iterate 25 times MP3 use of IF to check if the array element is less than the current lowest value MP4 ...if it is, set this to the lowest value MP5 output the result (with an appropriate message) after the loop Example answer: <pre> Min ← 100 FOR Count ← 1 TO 25 IF Temperatures[Count] < Min THEN Min ← Temperatures[Count] ENDIF NEXT Count OUTPUT "The lowest temperature is ", Min </pre>	4

Question	Answer	Marks
4(a)	One mark per mark point, max four MP1 Line 01 / DECLARE City ARRAY[1:50, 1:2] OF BOOLEAN should be DECLARE City : ARRAY[1:50, 1:2] OF STRING Line 05 / IF should be REPEAT MP2 Line 07 / INPUT City[Count, 2] should be INPUT City[Count, 1] MP3 Line 11 / UNTIL Count = 50 // Line 04 / Count ← 1 AND Line 10 / Count ← Count + 1 should be UNTIL Count = 51 / UNTIL Count > 50 // Line 04 / Count ← 0 AND move Line 10 to beginning of loop / Line 06 MP4 Line 12 / FOR Out ← 1 TO 1 should be FOR Out ← 1 TO 50 Correct algorithm: <pre> 01 DECLARE City : ARRAY[1:50, 1:2] OF STRING 02 DECLARE Count : INTEGER 03 DECLARE Out : INTEGER 04 Count ← 1 05 REPEAT 06 OUTPUT "Enter the name of the city" 07 INPUT City[Count, 1] 08 OUTPUT "Enter the name of the country" 09 INPUT City[Count, 2] 10 Count ← Count + 1 11 UNTIL Count > 50 12 FOR Out ← 1 TO 50 13 OUTPUT "The city ", City[Out, 1], " is in ", City[Out, 2] 14 NEXT Out </pre>	4

Question	Answer	Marks
4(b)	One mark per mark point, max five MP1 add an input (and prompt to ask) for the country to be searched MP2 ...between lines 11 and 12 MP3 ...using a new variable for the input MP4 Add an IF statement to check if the current Country array element matches the country being searched MP5 ...between lines 12 and 13 MP6 ...if it does, allow the output in line 13 // the output in line 13 should be after a THEN MP7 If it does not, check the next element.	5

Question	Answer	Marks																																																																																															
6(a)	One mark per correct column, max five <table><tr><th>Value</th><th>Average</th><th>Total</th><th>Count</th><th>OUTPUT</th></tr><tr><td></td><td></td><td>0</td><td>0</td><td></td></tr><tr><td>25</td><td></td><td>25</td><td>1</td><td></td></tr><tr><td>35</td><td></td><td>60</td><td>2</td><td></td></tr><tr><td>3</td><td></td><td>63</td><td>3</td><td></td></tr><tr><td>0</td><td>21</td><td></td><td></td><td>Total is 63</td></tr><tr><td></td><td></td><td></td><td></td><td>Average is 21</td></tr><tr><td></td><td></td><td>0</td><td>0</td><td></td></tr><tr><td>57</td><td></td><td>57</td><td>1</td><td></td></tr><tr><td>20</td><td></td><td>77</td><td>2</td><td></td></tr><tr><td>25</td><td></td><td>102</td><td>3</td><td></td></tr><tr><td>18</td><td></td><td>120</td><td>4</td><td></td></tr><tr><td>0</td><td>30</td><td></td><td></td><td>Total is 120</td></tr><tr><td></td><td></td><td></td><td></td><td>Average is 30</td></tr><tr><td></td><td></td><td>0</td><td>0</td><td></td></tr><tr><td>-1</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>	Value	Average	Total	Count	OUTPUT			0	0		25		25	1		35		60	2		3		63	3		0	21			Total is 63					Average is 21			0	0		57		57	1		20		77	2		25		102	3		18		120	4		0	30			Total is 120					Average is 30			0	0		-1																				5
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6(b)	One mark per mark point, max two MP1 to add together / find the average of a batch of numbers MP2 the total and average are output (when the batch is complete/when 0 is entered) MP3 when 0 is entered a new batch is started.	2																																																																																															

Question	Answer	Marks
1	A	1

Question	Answer	Marks
3(a)	One mark for each correct line from description to pseudocode keyword Pseudocode description stores data in a file retrieves data from a file displays data on a screen enters data from a keyboard Pseudocode keyword OUTPUT WRITE READ OPEN INPUT	4
3(b)	One mark for each point (max two) - data is stored permanently - data can be moved to another computer - another copy of data can be made and stored//accessed elsewhere // backup copy	2

Question	Answer	Marks
4(a)	One mark for each point - type check - range check	2
4(b)	One mark for each point (max five) - use of loop for check - checking for whole number - checking for number greater than or equal to one ... and less than or equal to six - Appropriate error/reinput message - ability to reinput value Example: <pre>WHILE Seats < 1 OR Seats > 6 OR Seats <> ROUND(Seats, 0) DO OUTPUT "Please enter a valid number of seats " INPUT Seats ENDWHILE</pre>	5
4(c)	One mark for correct test data, one mark for corresponding reason Example: 7, abnormal data to show that this value would be rejected	2

Question	Answer	Marks
5(a)	One mark for each error identified and correction given - Line 06 Password should be NewPassword - Line 11 AND should be OR - Line 16 INPUT should be OUTPUT	3

Question	Answer	Marks
5(b)	Max six marks from: Max four from: one mark for data entry with message one mark for initialisation one mark for checking list // decision box comparing input with array one mark for updating // updating the two variables position and found one mark for loop control // second decision box one mark for setting new password to position in list one mark for outputs // two outputs Two marks: one mark for correct use of flow chart symbols one mark for correct use arrows and labels <pre> graph TD Start([START]) --> Output1[/OUTPUT "Enter your new password"/] Output1 --> Input[/INPUT NewPassword/] Input --> Init[Posn ← 1 Found ← FALSE] Init --> Decision1{IS NewPassword = OldList[Posn]?} Decision1 -- Yes --> FoundTrue[Found ← TRUE] FoundTrue --> Decision2{IS Found OR OldList[Posn] = "XXXX"?} Decision1 -- No --> PosnInc[Posn ← Posn + 1] PosnInc --> Decision2 Decision2 -- No --> Decision1 Decision2 -- Yes --> Decision3{IS Found?} Decision3 -- Yes --> Output2[/OUTPUT "Password has been used before"/] Output2 --> FoundTrue Decision3 -- No --> Output3[/OUTPUT "New password accepted"/] Output3 --> Update[OldList[Posn] ← NewPassword] Update --> Stop([STOP]) </pre>	6

Question	Answer					Marks
8(a)	Accept	Reject	PartOK	Error	OUTPUT	5
	0	0				
	1		Y			
	2		Y			
	3		Y			
		1	N			
	4		Y			
	5		Y			
	6		Y			
	7		Y			
		2	N			
	8		Y			
	9		Y			
	10		Y	20		
					Too many rejected 20% error	
One mark for each column						

Question	Answer	Marks
8(b)	One mark for each point max three • after the Input box // before the first decision box • insert a process box • to convert the input to upper case OR • change the first decision / add another decision box • to accept 'y' as well • by adding OR PartOK = 'y'	3